

Final Exam Part 2 – CSC 410 Parallel Computing

Fall 2024

Due: Dec 17 @ 11:59pm

Given the sequential code for the N-body problem (`nBody.c`), complete the following tasks.

Coding Task 1: Convert it to parallel using Pthread.

Coding Task 2: Convert it to parallel using OpenMP.

Coding Task 3: Convert it to parallel using MPI.

Coding Task 4: Convert it to parallel using OpenCL. Use the starter code `ocl_nbody.c`

Task 5: Benchmark the time for the sequential version and parallel versions - coding tasks 1 to 4.

Note: Use ***time ./a.out*** for the sequential version and the appropriate timing function for each parallel programming tool.

Task 6: Explain the following in detail:

- a) How did you partition the work with each tool.
- b) Explain the necessity of the operations you used in detail for each parallel programming tool.
- c) Compare and contrast each tool for parallelizing the N-body sequential program. List their advantages and disadvantages.

Total Points - 60

- Code runs and works as expected - Task 1: 10 points
- Code runs and works as expected - Task 2: 10 points
- Code runs and works as expected - Task 3: 10 points
- Code runs and works as expected - Task 4: 10 points
- Clear documentation - Task 5: 5 points

- Clear and detailed explanation - Task 6: 15 points
- Available on GitHub